VITyarthi | Open Source Software

The Open Source Report

A Capstone Report for OSS NGMC Course

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Introduction

This report presents an audit of **Git,** a widely used open-source version control system. The objective of this project is to understand the origin, licensing, ethical aspects, Linux implementation, and ecosystem of **Git**. It also compares **Git** with proprietary alternatives to evaluate its effectiveness in real-world applications.

Your choice	Git

Project Report

Part A — Origin and Philosophy (Units 1 & 2)

A1. The problem this software was created to solve

In 2005, Linux developers used a version control system called BitKeeper in order to manage their code. However, BitKeeper was not completely open source and after some licensing issues the Linux community lost access to it.

This created a serious issue because managing a large project without a proper version control system was extremely difficult. To solve this issue , Linus Torvalds creator Linux built a new system that would be fast, efficient and completely open source.
This marked the birth of Git with a focus on performance, data integrity and distributed development. Unlike traditional systems, Git allowed every developers to have a full copy of the project , making collaboration easier and more reliable.

Git was not created as a planes product but as a quick solution to a real world problem. Today it has become one oof the most important tools in software development.

A2. The license — what it actually says

Git is released under the GNU (General Public License version 2). It is one of the most commonly used licenses for open-source software. This license guarantees that the software is free and open for everyone.

The license is based on four essential freedoms.

The user can use the software for any purpose, study how the software works by looking at the source code, modify the software to suit their needs, and distribute the copies of the software.

One of the most interesting things about this license is that it follows a copyleft principle, meaning that if any individual modifies the source code of Git and distributes it, they are also obligated to distribute their modified source code.

If any company wants to use Git, they can use it for their purpose, but they can also modify it for their use without distributing the source code. However, if they distribute the source code to other people, they are obligated to do so.

This type of license is powerful because it maintains the freedom of the user, at the same time allowing them to contribute to the development of the software.

GPL vs. MIT License Comparison

Both GNU General Public License and MIT License are popular open-source licenses, but they are different from each other. The GNU General Public License is a copyleft license, i.e., if someone wants to make any modification in the software, then he/she will have to release it under the same license. The MIT License, on the other hand, is a permissive license, i.e.,

anyone can make modifications in the code and even use it for proprietary software without disclosing any information.

If I have to choose a license for my own software, I would go for the GNU General Public License.

A3. The ethics of open source

Open source software is not just technical concept but also an ethical idea about how knowledge should be shared. It promotes collaboration. Transparency, and equal access to technology.

It allows anyone to use and improve software without any kind of restriction. This leads to faster innovation and helps students, developers, and organizations who may not have the resources to buy expensive proprietary tools. It also increases transparency because anyone can inspect the code for security issues.

But, not all software can or should be open source. Companies invest significant time and money into developing software and they may need to keep it proprietary to recover their costs and maintain a competitive advantage. In some cases, open source tools sourcing everything may reduce incentives for private innovation.

Many companies build successful products using open – source tools like Linux and Git. While this is not inherently wrong, it raises concerns about fairness if companies benefit from community contributions without giving back. Ideally, companies should contribute to the projects they depend on, either through code , funding or support.

Part B — Linux Footprint (Unit 2)

Git installation is very simple and straightforward. It can be installed easily using the package manager available in the Linux system. For Debian-based systems, the installation command is sudo apt install git. This makes the installation of the Git system accessible to the user.

Once the installation is done, the installation is checked using the command git version. This shows the version of the installed Git system. This is a confirmation that the installation is done properly.

The location of the installed Git system is checked using the which git command. This shows the location of the installed Git system, which is /usr/bin/git. This is the default location in the Linux system where user-level programs are stored.

The whereis git command is also available, which shows more detailed information about the files related to the installed Git system.

Unlike the server-based system, the Git system is not installed to run in the background. It is designed to run only when the user runs the required command, such as git init, git clone, or git commit.

The user is the one who runs the system, and the user is the user account created in the system. This is the user who runs the system, which is different from the user account created in the system.

The updates are taken care of by the open-source community, and the updates are available through the package manager of the system.

Part C — The FOSS Ecosystem (Units 3 & 4)

Git, however, does not work in isolation. It works in collaboration with a number of tools and platforms that are part of a larger open source ecosystem.

Git works hand in hand with a number of fundamental library and system tools that are available in Linux. This includes file systems, shell environments, and standard tools. This allows Git to perform a number of operations efficiently.

On the other hand, Git has led to the creation of a number of very powerful tools and platforms. For example, platforms such as GitHub and GitLab are all based on Git. The platforms offer a number of features that are useful for developers.

Git plays a very important role in web development and software engineering. It works well in a number of environments, for example, the LAMP environment, where Linux, Apache, MySQL, and PHP are used. In this environment, Git is very useful for developers.

Git has a very large number of users. A number of developers from all over the world are working to improve the functionality of the platform. The developers communicate through a number of mailing lists and online forums. The platform is maintained by a number of developers, led by Linus Torvalds.

In my opinion, Git is a very good example of how open source software evolves and grows. It not only solves a problem but also allows developers to create a number of tools and platforms.

Part D — Open Source vs Proprietary (Critical Analysis)

Dimension	Open Source	Proprietary Alternative
Cost	Free to use	Usually paid or licensed
Security (who can audit the code?)	Code is open, anyone can audit it	Code is closed , only company can review
Support and reliability	Community support + documentation	Official company support
Freedom to modify	Full freedom to modify and distribute	Very limited or no modification allowed
Community vs corporate control	Community- driven	Company-controlled

From this comparison, I would highly recommend the use of Git in real-world scenarios. It is highly reliable and efficient and widely used in the software industry. It is also highly flexible because it is a distributed system that allows developers to work individually without losing synchronization with the original project. Also, since it is open-source software, it is transparent and highly secured.

Although proprietary software might be better in terms of customer service and ease of use for novice developers, it is highly limited in flexibility and freedom. I personally believe that I do not only plan to use Git in the future but also participate in open-source software development in the future.

Shell Script Tasks

Script 1 — System Identity Report (Unit 1 & 2)

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Ishani Sahay

STUDENT_NAME="Ishani Sahay "
SOFTWARE_CHOICE="Git"

KERNEL=$(uname -r)
USER_NAME=$(whoami)
UPTIME=$(uptime -p)
DATE=$(date)
DISTRO=$(lsb_release -d | cut -f2)

echo "========================================"
echo " Open Source Audit — $STUDENT_NAME"
echo "========================================"
echo "Software : $SOFTWARE_CHOICE"
echo "Kernel   : $KERNEL"
echo "User     : $USER_NAME"
echo "Uptime   : $UPTIME"
echo "Date     : $DATE"
echo "Distro   : $DISTRO"
echo "License  : GPL (Open Source License)"
```

```
========================================
 Open Source Audit – Ishani Sahay
========================================
Software : Git
Kernel   : 3.5.7-463ebcdc.x86_64
User     : ishan
Uptime   :
Date     : Mon Mar 30 20:16:34 IST 2026
Distro   :
License  : GPL (Open Source License)

ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$
```

Script 2 — FOSS Package Inspector (Unit 2)

Starter structure

```
#!/bin/bash
# Script 2: FOSS Package Inspector

PACKAGE="git"

if dpkg -l | grep -q $PACKAGE; then
    echo "$PACKAGE is installed."
    git --version
else
    echo "$PACKAGE is NOT installed."
fi

case $PACKAGE in
    git) echo "Git: Version control system that
powers modern development" ;;
    apache2) echo "Apache: Web server for hosting
websites" ;;
    mysql-server) echo "MySQL: Database system"
;;
    vlc) echo "VLC: Media player" ;;
esac
```

```
ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$ nano script2.sh

ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$ ./script2.sh
git is installed.
git version 2.49.0.windows.1
Git: Version control system that powers modern development
```

Script 3 — Disk and Permission Auditor (Unit 2)

Starter structure

```
#!/bin/bash
# Script 3: Disk and Permission Auditor

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------"

for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')
```

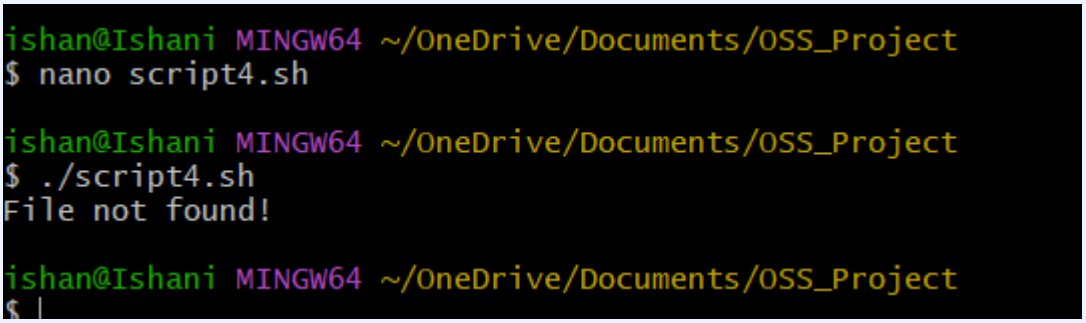
```
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
        echo "$DIR => Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist"
    fi
done

if [ -f "$HOME/.gitconfig" ]; then
    echo "Git config exists"
else
    echo "Git config not found"
fi
```

```
ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$ nano script3.sh

ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$ ./script3.sh
Directory Audit Report
----------------------
/etc => Permissions: drwxr-xr-x ishan 197609 | Size: 1.5M
/var/log does not exist
/home does not exist
/usr/bin => Permissions: drwxr-xr-x ishan 197609 | Size: 104M
/tmp => Permissions: drwxr-xr-x ishan 197609 | Size: 225M
Git config exists

ishan@Ishani MINGW64 ~/OneDrive/Documents/OSS_Project
$
```

Script 4 — Log File Analyzer (Units 2 & 5)

Starter structure

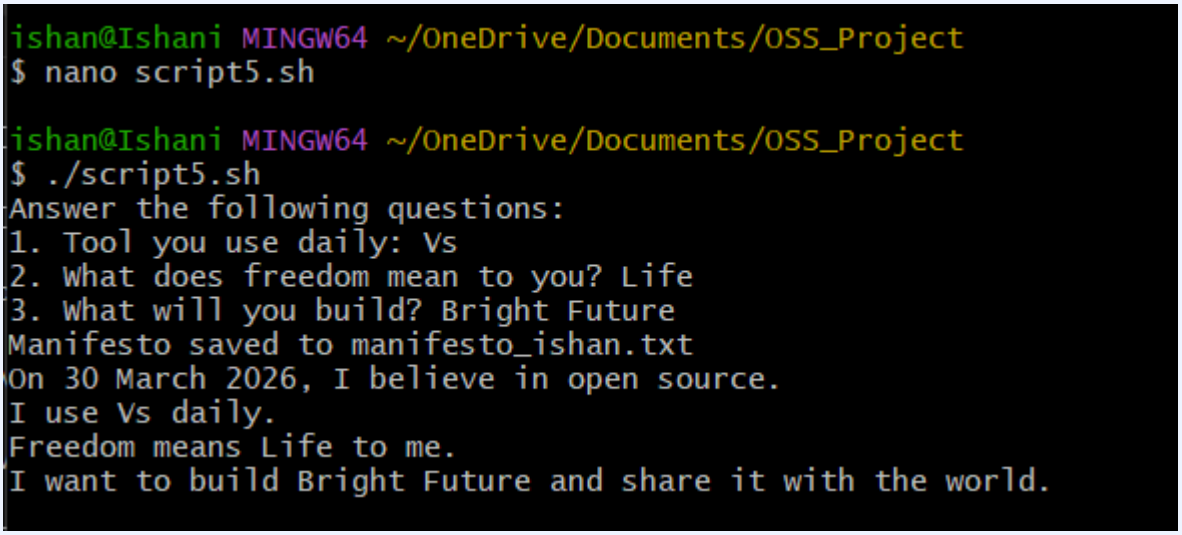
```
#!/bin/bash
# Script 4: Log File Analyzer

LOGFILE=$1
KEYWORD=${2:-"error"}
COUNT=0

if [ ! -f "$LOGFILE" ]; then
    echo "File not found!"
    exit 1
fi

while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times"

echo "Last 5 matching lines:"
grep -i "$KEYWORD" "$LOGFILE" | tail -5
```



Script 5 — The Open Source Manifesto Generator (Unit 5)

Starter structure

```
#!/bin/bash
# Script 5: Open Source Manifesto Generator

echo "Answer the following questions:"

read -p "1. Tool you use daily: " TOOL
read -p "2. What does freedom mean to you? " FREEDOM
read -p "3. What will you build? " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

echo "On $DATE, I believe in open source." > $OUTPUT
echo "I use $TOOL daily." >> $OUTPUT
echo "Freedom means $FREEDOM to me." >> $OUTPUT
echo "I want to build $BUILD and share it with the world." >> $OUTPUT

echo "Manifesto saved to $OUTPUT"
cat $OUTPUT
```



Conclusion

This project has helped me gain a good understanding of how open source systems actually work. I became familiar with the fundamental concepts behind the open source philosophy, the structure of the Linux system, and the basics of shell scripting.

I think that the experience of using tools like Git has shown me how real-world development is done. It is not just about developing code on your own; it is about developing something together.

I think that I now understand that Git is not just another tool but is actually part of the process of modern software development.

References

- GNU Official Website
- Git Documentation
- Linux Manual Pages